# VIKUM BANDARA

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## Summary

Mathematics researcher and university instructor working in combinatorics, discrete dynamical systems, graph theory, and cryptography. Uses Python to explore examples, visualize structures, and test conjectures. Coauthor of two journal articles on cryptography and an arXiv preprint on generalizations of Bulgarian solitaire, submitted for publication.

## Education and Training

**Clemson University** Clemson, SC, USA
Master of Science in Mathematical Sciences December 2025
Research focus: combinatorics and discrete dynamical systems.
Cumulative graduate GPA: 3.57/4.00 (through Spring 2026).
Passed the Operations Research preliminary examination for Ph.D. candidacy.
Relevant coursework: Probability, Stochastic Processes, Data Analysis, Matrix Analysis, Mathematical Programming, Advanced Linear Programming, Discrete Optimization, Network Flow Programming, Abstract Algebra, Introduction to Dynamical Systems Theory, Combinatorial Analysis, Cryptography, and Theory of Graphs.

**University of Peradeniya** Peradeniya, Sri Lanka
Bachelor of Science (Hons) in Mathematics September 2022
Major: mathematics (pure and applied); minor: physics.
Mathematics GPA: 3.77/4.00; overall GPA: 3.66/4.00. Four-year degree.

**G.C.E. Advanced Level (Physical Science)** Sri Lanka
Mathematics: A; Physics: B; Chemistry: C December 2016

**IELTS Academic (2022)**
Overall: 7.0 (Reading: 8.0; Writing: 6.5; Listening: 7.0; Speaking: 6.5).

## Research Interests

Algebraic and combinatorial structures in discrete dynamical systems; orbit structure and permutation actions; labeled configurations; enumerative combinatorics; graph theory; cryptography; number theory; computational exploration.

## Publications and Preprints

**V. Bandara** and S. Poznanović, *Bulgarian solitaire with distinguishable cells*. Preprint, arXiv:2611.10437 (2026); submitted for publication.

R. Ranasinghe and **V. Bandara**, *Collatz conjecture–driven cryptographic framework*. Journal of Discrete Mathematical Sciences & Cryptography, 29(7), 2621–2640, 2026. doi:10.47974/JDMSC-2232.

P. G. R. S. Ranasinghe, **R. M. V. V. Bandara**, and A. M. S. L. K. Athapaththu, *Symmetric encryption using snake graphs and supermagic covering*. Journal of the National Science Foundation of Sri Lanka, 52(4), 435–440, 2024. doi:10.4038/jnsfsr.v52i4.12196.

**V. Bandara** and R. Ranasinghe, *Novel Cryptographic Scheme Based on the Collatz Conjecture.* RESCON Research Conference, 2022.

## Research

**Research Project in Combinatorics**, Clemson University
Master's project (PDF)

- Investigated generalizations of Bulgarian solitaire with Dr. Svetlana Poznanović.
- Used Python to visualize orbits and observed that a configuration can return to its original shape before individual cards return to their original positions.
- Developed a coloring algorithm on Ferrers diagrams that connects the movement of labeled cells to permutation cycles and orbit sizes.
- Coauthored *Bulgarian solitaire with distinguishable cells*, available on arXiv and submitted for publication.

**Final-Year Research Project**, University of Peradeniya
Undergraduate thesis (PDF)

- Studied relationships among Collatz sequences with Dr. Rajitha Ranasinghe.
- Implemented a proposed cryptographic scheme in Python using the behavior of Collatz sequences.
- Coauthored the resulting paper, *Collatz conjecture–driven cryptographic framework* (2026).

**Additional Research Projects**, University of Peradeniya

- **Cryptographic Scheme Based on Graph Theory** (October 2022–2023): Studied snake vertex labelings and supermagic coverings of wheel graphs for symmetric encryption. This work led to *Symmetric encryption using snake graphs and supermagic covering*, coauthored with P. G. R. S. Ranasinghe and A. M. S. L. K. Athapaththu (2024).
- **Constructive Approach to the Collatz Conjecture** (June 2021–2023): Constructed a framework for comparing Collatz sequences and studied its behavior on odd positive integers.
- **Novel Cryptographic Scheme Based on the Collatz Conjecture** (June 2021–2023): Developed a symmetric encryption and decryption algorithm based on Collatz sequences. Presented this work at RESCON 2022.

## Experience

**Graduate Teacher of Record**, Clemson University
January 2025–Present
MATH 1020 – Business Calculus I

**Graduate Teaching Assistant**, Clemson University
August 2023–December 2024
MATH 1040 – Precalculus and Introductory Differential Calculus; MATH 1080 – Calculus of One Variable II; MATH 2080 – Introduction to Ordinary Differential Equations; MATH 3110 – Linear Algebra

**Teaching Assistant**, University of Peradeniya
September 2022–April 2023

**Venue Coordinator and Presenter**, 2022
SURS Science Undergraduate Research Symposium

**Academic Coordinator and Presenter**, 2022
RESCON Research Conference

## Skills

**Programming:** Python (computational experiments, visualization, and cryptographic implementation).
**Familiar with:** MATLAB, Julia, Gurobi, RStudio.
**Technical writing:** LaTeX.
**Languages:** English and Sinhala.

## References

**Dr. P. G. R. S. Ranasinghe**,
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**Dr. Svetlana Poznanović**,
Professor and Associate Director for Mathematics, School of Mathematical and Statistical Sciences, Clemson University, SC 29634, USA.
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